SHILPA S

THIRUVANANTHAPURAM, KERALA, INDIA

9495252341 shilpamurukan00@gmail.com linkedin.com/in/shilpa-s-stat Portfolio Website

Profile Summary

MSc Statistics graduate with a multidisciplinary background in statistical analysis, data analytics, research, and teaching. Experienced in developing interactive dashboards, analyzing complex datasets, applying statistical methods, and communicating analytical findings through academic and practical projects. Proficient in Python, R, SQL, Excel, Power BI, and Tableau, with a strong foundation in statistical modelling, forecasting, data visualization, and evidence-based decision-making. Combines quantitative reasoning, analytical problem-solving, and communication skills with a commitment to continuous learning and professional development.

Skills

Statistics

- Time Series Forecasting
- Statistical Modelling
- Machine Learning
- Regression Analysis

Programming Languages

- Python
- SQL
- R

Analytical Tools

- Power BI
- Tableau
- Advanced Excel

Experience

Sree Narayana College, Chempazhanthy **July 2026 – Present**
Guest Lecturer - Statistics (Department of Mathematics) *Kerala, India*

- Delivered undergraduate lectures on statistical concepts, adapting instructional approaches to students from diverse academic disciplines.
- Developed teaching materials and structured learning activities to facilitate understanding of statistical principles.
- Prepared assignments and assessments to evaluate student learning and conceptual understanding.

ICT Academy of Kerala (ICTAK) **Sept. 2025 – Jan. 2026**
Research and IT Solutions Intern *Kerala, India*

- Implemented training analytics dashboards using Power BI and Python covering 180+ programs and 27 trainers
- Designed 10+ KPIs to track learner progress and faculty performance
- Cleaned, transformed and modeled datasets using Power Query
- Created interactive visual reports to support training performance monitoring and decision making
- **Tools:** Power BI, Python, Excel, Power Query

Projects

Rice Yield Time Series Forecasting Project **2025**
R, Time Series Analysis

- Processed 22 years of seasonal crop yield data to identify production trends
- Developed ARIMA forecasting model reducing prediction error by 19%
- Evaluated models using RMSE, MAE, MAPE and Theil's U
- Detected 2018 flood anomaly using outlier detection
- **Tools:** R, ARIMA, Time Series Analysis
- **Project Repository:** GitHub

Mental Health Analysis during covid 19 **2021**
Python, Statistical Analysis

- Collected primary data through a structured Google Forms questionnaire to study mental health trends during the COVID-19 pandemic.
- Identified key factors associated with mental health during the pandemic.
- Developed visualizations to highlight trends and patterns in the survey data.
- **Tools:** Python, Pandas, NumPy, Statistical Testing
- **Project Repository:** GitHub

Inventory and Supply Chain Analytics Dashboard **2026**

Power BI, Data Visualization

- Developed Power BI dashboard tracking inventory KPIs
- Evaluated inventory turnover and lead time metrics
- Developed KPI cards and interactive filters
- Enabled regional performance comparison
- **Tools:** Power BI, Excel
- **Project Repository:** GitHub

Virtual Experience

British Airways Data Science Virtual Experience **March-2026**

Remote

- Trained a Random Forest classification model on 50K+ customer records achieving ∼85% prediction accuracy
- Executed data preprocessing and feature engineering on 10+ variables
- Identified key booking drivers such as purchase lead time, route and length of stay using feature importance analysis
- Conducted 5-fold cross validation and generated insights to support customer conversion strategies
- **Tools:** Python, Machine Learning, Random Forest, Data Preprocessing

Lloyds Banking Group Data Science Virtual Experience **March-2026**

Remote

- Examined and integrated banking datasets and performed exploratory data analysis to identify customer churn indicators
- Developed and compared Logistic Regression and Random Forest models achieving 78% prediction accuracy
- Improved model robustness using cross validation and hyperparameter tuning
- Engineered 10+ predictive features and evaluated models using ROC-AUC, precision and recall
- Generated insights on key churn drivers to support customer retention strategies
- **Tools:** Python, Machine Learning, Logistic Regression, Random Forest, EDA

Education

Master of Science in Statistics **2023 – 2025**

University of Kerala - 84% *University College,TVM*

B.Sc in Statistics **2018 - 2021**

University of Kerala - 87% *Govt College Kariavattom, TVM*

Certifications

Microsoft Power BI Data Analyst Professional Certificate - Microsoft via Coursera (2026)

Certified Specialist in Data Science and Analytics - ICT Academy of Kerala (2025)

Tableau 10 Essential Training - LinkedIn Learning (2025)